



Problems (Matrices)

1. What is the condition on a , b , c such that the following linear system of equations have a unique solution

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c$$

2. Calculate the minors and co-factors of $A = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 0 & 1 \\ 3 & 2 & 1 \end{pmatrix}$

3. Determine the values of t for which $|A| = 0$ where

$$A = \begin{pmatrix} t-2 & 4 & 3 \\ 1 & t+1 & -2 \\ 0 & 0 & t-4 \end{pmatrix}$$

4. Find the matrix P that projects every vector b in $\mathbb{R}^3$ onto the line in the direction of $a = (2, 1, 3)$. What are the column space and nullspace of P ? Describe them geometrically and give a basis for each space. What are all the eigenvectors of P and their corresponding eigenvalues?
5. Circle the words that correctly complete the following sentence: (a) Suppose a 3 by 5 matrix A has rank $r = 3$. Then the equation $Ax = b$ (always / sometimes but not always) has (a unique solution / many solutions / no solution). (b) What is the column space of A ? Describe the nullspace of A .